```
#include <LiquidCrystal.h>
#include <SoftwareSerial.h>

// LCD pins same as you gave
LiquidCrystal lcd(12, 11, 4,5,6,7,8,9,10,2);

// SIM800 pins A2 = RX, A3 = TX
SoftwareSerial sim( A2, A3 ); // RX, TX

const int vibPin = 3;
const int redLED = 13;
const int blueLED = A0;
const int buzzerPin = A1;

bool sentAlready = false; // to avoid multiple SMS

void setup() {
  pinMode(vibPin, INPUT);
  pinMode(redLED, OUTPUT);
  pinMode(blueLED, OUTPUT);
  pinMode(buzzerPin, OUTPUT);

  lcd.begin(16,2);
  lcd.clear();
  lcd.print("System Ready");

  sim.begin(9600);
  delay(3000);
}

void sendSMS()
{
  String msg = "EARTHQUAKE ALERTED";

  const char* numbers[5] = {
    "+919060776677",
    "+919743235699",
    "+918884759639",
    "+919353802953",
    "+919844198011"
  };

  for(int i=0;i<5;i++)
  {
    sim.print("AT+CMGF=1\r");
    delay(200);
    sim.print("AT+CMGS=\"");
    sim.print(numbers[i]);
    sim.print("\"\r");
    delay(200);
    sim.print(msg);
    delay(200);
    sim.write(26); // CTRL+Z
    delay(2000); // 2 sec minimum
```

```
  }
}

void loop() {

  int vib = digitalRead(vibPin);

  if(vib == HIGH) {
    digitalWrite(redLED, HIGH);
    digitalWrite(blueLED, LOW);
    digitalWrite(buzzerPin, HIGH);
    lcd.clear();
    lcd.print("EARTHQUAKE !!");

    if(!sentAlready) {
      sendSMS();
      sentAlready = true;
    }
  }
  else {
    digitalWrite(redLED, LOW);
    digitalWrite(blueLED, HIGH);
    digitalWrite(buzzerPin, LOW);
    lcd.clear();
    lcd.print("No vibration");
  }

  delay(5);  // smallest safe delay
}
```